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1. 习题三：

1.

X 的可能取值为 0, 1, 2, 3, Y 的可能取值为 0, 1, 2

则 X, Y 的联合分布律为：

$$P(x = i, Y = j) = P(X = i)P(Y = j|X = i) = \frac{C_3^i C_4^{4-i}}{C_7^4} \times \frac{C_2^j C_2^{4-i-j}}{C_4^{4-i}} = \frac{C_3^i C_2^j C_2^{4-i-j}}{35}, i = 0, 1, 2, 3; j = 2 - i, \dots 4 - i$$

p_{ij}		X				$p_{\cdot j}$ $= \sum_i p_{ij}$
		0	1	2	3	
Y	0	0	0	$\frac{3}{35}$	$\frac{2}{35}$	$\frac{5}{35}$
	1	0	$\frac{6}{35}$	$\frac{12}{35}$	$\frac{2}{35}$	$\frac{20}{35}$
	2	$\frac{1}{35}$	$\frac{6}{35}$	$\frac{3}{35}$	0	$\frac{10}{35}$
$p_{i\cdot} = \sum_j p_{ij}$		$\frac{1}{35}$	$\frac{12}{35}$	$\frac{18}{35}$	$\frac{4}{35}$	1

3.

(1)X,Y 的可能取值均为 1,2,...,n, (X, Y) 的联合分布律为

$$P(X = i, Y = j) = P(X = i)P(Y = j|X = i) = \begin{cases} \frac{1}{n} \times \frac{1}{n-1} = \frac{1}{n(n-1)}, & j \neq i \\ 0, & i = j \end{cases} \quad i = 1, 2, \dots, n; j = 1, 2, \dots, n.$$

(2) 当 n=3 时, 由 (1) 可知

$$P(X = i, Y = j) = \frac{1}{6}$$

故 (X, Y) 的联合分布律为

P_{ij}		X			$p_{\cdot j} = \sum_i p_{ij}$
		1	2	3	
Y	1	0	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{3}$
	2	$\frac{1}{6}$	0	$\frac{1}{6}$	$\frac{1}{3}$
	3	$\frac{1}{6}$	$\frac{1}{6}$	0	$\frac{1}{3}$
$p_{i\cdot} = \sum_j p_{ij}$		$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	1

4.

由题意得 X 的分布函数为

$$F(x) = \begin{cases} 0, & x < -1 \\ \frac{x+1}{3}, & -1 \leq x \leq 2 \\ 1, & x > 2 \end{cases}$$

$$P(Y_1 = 0, Y_2 = -1) = 0$$

$$P(Y_1 = 1, Y_2 = -1) = P(0 < X < 1) = F(1) - F(0) = \frac{1}{3}$$

$$P(Y_1 = 2, Y_2 = -1) = P(X \geq 1) = 1 - F(1) = \frac{1}{3}$$

$$P(Y_1 = 0, Y_2 = 1) = P(X < 0) = F(0) = \frac{1}{3}$$

$$P(Y_1 = 1, Y_2 = 1) = P(X = 0) = 0$$

$$P(Y_1 = 2, Y_2 = 1) = 0$$

故 (Y_1, Y_2) 的联合分布律为

P_{ij}		Y_1			$p_{\cdot j} = \sum_i p_{ij}$
		0	1	2	
Y_2	-1	0	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{2}{3}$
	1	$\frac{1}{3}$	0	0	$\frac{1}{3}$
$p_{i\cdot} = \sum_j p_{ij}$		$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	1

边缘分布律为

Y_1	0	1	2
P	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$

Y_2	-1	1
P	$\frac{2}{3}$	$\frac{1}{3}$

5.

$$(1) \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = \frac{k}{8} = 1$$

$$\therefore k = 8$$

$$(2) P(0 \leq X \leq 2, 0 < Y \leq 1) = \int_0^2 (\int_0^1 f(x, y) dy) dx = (1 - e^{-4})^2$$

$$(3) P(X + Y < 1) = \int_0^1 \int_0^{1-x} 8e^{-2x-4y} dy dx = (1 - e^{-2})^2$$

$$(4) F(x, y) = \int_{-\infty}^x \int_{-\infty}^y f(u, v) du dv = \begin{cases} \int_0^x \int_0^y f(u, v) du dv, & x > 0, y > 0 \\ 0, & \text{others} \end{cases}$$

$$= \begin{cases} 1 - e^{-2x} - e^{-4y} + e^{-2x-4y}, & x > 0, y > 0 \\ 0, & \text{others} \end{cases}$$

4 7.

$$(1) \iint_G f(x, y) = \int_0^2 \int_0^{\frac{x}{2}} f(x, y) dy dx = \frac{4}{3} k$$

$$\therefore k = \frac{3}{4}$$

$$(2) P(X + Y \leq 2) = \int_0^{\frac{4}{3}} \int_0^{\frac{x}{2}} f(x, y) dy dx + \int_{\frac{4}{3}}^2 \int_0^{2-x} f(x, y) dy dx = \frac{5}{9}$$

$$(3) f_X(x) = \begin{cases} \int_0^{\frac{x}{2}} f(x, y) dy, & 0 < x < 2 \\ 0, & \text{others} \end{cases} = \begin{cases} \frac{3}{8} x^2, & 0 < x < 2 \\ 0, & \text{others} \end{cases}$$

$$f_Y(y) = \begin{cases} \int_{2y}^2 f(x, y) dx, & 0 < y < 1 \\ 0, & \text{others} \end{cases} = \begin{cases} \frac{3}{2} - \frac{3}{2} y^2, & 0 < y < 1 \\ 0, & \text{others} \end{cases}$$

8.

$$(1) f_X(x) = \begin{cases} \int_{x^2}^1 f(x, y) dy, & -1 < x < 1 \\ 0, & \text{others} \end{cases} = \begin{cases} \frac{21}{8} (x^2 - x^4), & 0 < x < 2 \\ 0, & \text{others} \end{cases}$$

$$f_Y(y) = \begin{cases} \int_{-\sqrt{y}}^{\sqrt{y}} f(x, y) dx, & 0 < y < 1 \\ 0, & \text{others} \end{cases} = \begin{cases} \frac{7}{2} y^{\frac{5}{2}}, & 0 < y < 1 \\ 0, & \text{others} \end{cases}$$

$$(2) P(X \geq Y) = \int_0^1 \int_{x^2}^x f(x, y) dy dx = \frac{3}{20}$$

9.

(1) $\therefore (X, Y)$ 服从均匀分布

$$\therefore S_G = \int_0^1 (x^2 - \frac{1}{2} x^2) dx = \frac{1}{6}$$

$$\therefore (X, Y) \text{ 的联合概率密度为 } f(x, y) = \begin{cases} 6, & (x, y) \in G \\ 0, & \text{others} \end{cases}$$

$$(2) f_X(x) = \begin{cases} \int_{\frac{x^2}{2}}^{x^2} f(x, y) dy, & 0 < x < 1 \\ 0, & \text{others} \end{cases} = \begin{cases} 3x^2, & 0 < x < 1 \\ 0, & \text{others} \end{cases}$$

$$f_Y(y) = \begin{cases} \int_{\sqrt{y}}^{\sqrt{2y}} f(x,y)dx, & 0 < y < \frac{1}{2} \\ \int_{\sqrt{y}}^1 f(x,y)dx, & \frac{1}{2} < y < 1 \\ 0, & \text{others} \end{cases} = \begin{cases} 6\sqrt{2y} - 6\sqrt{y}, & 0 < y < \frac{1}{2} \\ 6 - 6\sqrt{y}, & \frac{1}{2} < y < 1 \\ 0, & \text{others} \end{cases}$$

14.

$$(1) f_X(x) = \int_{-\infty}^{+\infty} f(x,y)dy = \begin{cases} \int_0^{+\infty} \frac{x^3}{2} e^{-x(1+y)} dy = \frac{x^2}{2} e^{-x}, & x > 0 \\ 0, & \text{others} \end{cases}$$

(2) 当 $x > 0$ 时,

$$f_{Y|X}(y|x) = \begin{cases} \frac{f(x,y)}{f_X(x,y)} = x e^{-xy}, & y > 0 \\ 0, & \text{others} \end{cases}$$

$$f_{Y|X}(y|0.5) = \begin{cases} 0.5 e^{-0.5y}, & y > 0 \\ 0, & y \leq 0 \end{cases}$$

$$(3) P(Y \geq 1 | X = 0.5) = \int_1^{+\infty} f_{Y|X}(y|0.5)dy = \int_1^{+\infty} 0.5 e^{-0.5y} dy = e^{-0.5}$$

16.

$$(1) f_X(x) = \int_{-\infty}^{+\infty} f(x,y)dy = \int_0^1 \frac{2}{5}(2x+3y)dy = 0.8x + 0.6, 0 \leq x \leq 1$$

$$P(X \geq 0.8) = \int_{0.8}^1 f_X(x)dx = 0.264$$

(2) 当 $0 < Y < 1$ 时, X 有条件概率密度, 且

$$f_{X|Y}(x|y) = \begin{cases} \frac{f(x,y)}{f_Y(y)}, & 0 < x < 1 \\ 0, & \text{others} \end{cases}$$

$$\therefore f_Y(y) = \begin{cases} \int_0^1 \frac{2}{5}(2x+3y)dx = 1.2y + 0.4, & 0 < y < 1 \\ 0, & \text{others} \end{cases}$$

$$\therefore f_{X|Y}(x|y) = \begin{cases} \frac{2x+3y}{3y+1}, & 0 < x < 1 \\ 0, & \text{others} \end{cases}$$

$$(3) P(0.6 \leq X \leq 0.8 | Y = 0.3) = \int_{0.6}^{0.8} f_{X|Y}(x|0.3)dx = \int_{0.6}^{0.8} \frac{2x+0.9}{1.9} dx \approx 0.242$$

23.

$$f_X(x) = \begin{cases} \int_0^1 f(x, y) dy = \int_0^1 (x + y) dy = x + 0.5, & 0 < x < 1 \\ 0, & \text{others} \end{cases}$$

$$P(0 < X < \frac{1}{n}) = \int_0^{\frac{1}{n}} f_X(x) = \frac{1}{2n^2} + \frac{1}{2n}$$

$$P(0 < X < \frac{1}{n}, Y \leq y) = \begin{cases} 0, & y < 0 \\ \int_0^{\frac{1}{n}} \int_0^y f(x, y) dy dx = \frac{y}{2n^2} + \frac{y^2}{2n}, & 0 \leq y \leq 1 \\ \frac{1}{2n^2} + \frac{1}{2n}, & y > 1 \end{cases}$$

$$\therefore F_{Y|X}(y|0 < x < \frac{1}{n}) = \frac{P(0 < X < \frac{1}{n}, Y \leq y)}{P(0 < X < \frac{1}{n})} = \begin{cases} 0, & y < 0 \\ \frac{ny^2 + y}{n+1}, & 0 \leq y \leq 1 \\ 1, & y > 1 \end{cases}$$

2. 3.1-6 补充题一:

$$(1) \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dy dx = \int_0^1 \int_0^1 cxy(x + y) dy dx = \frac{c}{3} = 1$$

$$\therefore c = 3$$

$$(2) P(X + Y \geq 1) = \int_0^1 \int_{1-x}^1 3xy(x + y) dy dx = 0.9$$

$$(3) f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \int_0^1 3xy(x + y) dy = \frac{3}{2}x^2 + x$$

3.1-6 补充题二:

$$(1) \therefore S_G = \frac{1}{2}$$

$$\therefore f(x, y) = \begin{cases} 2, & (x, y) \in G \\ 0, & \text{otherwise} \end{cases}$$

$$(2) P(Y \geq X^2) = \int_0^1 \int_{x^2}^x f(x, y) dy dx = \int_0^1 (x - x^2) dx = \frac{1}{6}$$

3.2-6 补充题一:

$$f(x, y) = f_X(x)f_{Y|X}(y|x) = \begin{cases} 8xy, & 0 \leq x \leq 1, x \leq y \leq 1 \\ 0, & \text{others} \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y)dx = \begin{cases} \int_0^y 8xydx = 4y^3, & 0 \leq y \leq 1 \\ 0, & \text{others} \end{cases}$$

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{2x}{y^2}, 0 \leq y \leq 1, 0 \leq x \leq y$$

$$P(X > \frac{1}{2} | Y = \frac{2}{3}) = \int_{\frac{1}{2}}^{\frac{2}{3}} f_{X|Y}(X > \frac{1}{2} | Y = \frac{2}{3})dx = \int_{\frac{1}{2}}^{\frac{2}{3}} \frac{9}{2}x dx = \frac{7}{16}$$

3.2-6 补充题二:

$$(1) \int_{-\infty}^{+\infty} f(x, y)dydx = \frac{c}{3} = 1$$

$$\therefore c = 3$$

$$P(X \leq Y) = \int_0^1 \int_0^y f(x, y)dx dy = 0.5$$

$$P(X \leq 0.3, X \leq Y) = P(X \leq Y) - P(X > 0.3, X \leq Y) = P(X \leq Y) - \int_{0.3}^1 \int_{0.3}^y f(x, y)dx dy = 0.0585$$

$$P(X \leq 0.3 | X \leq Y) = \frac{P(X \leq 0.3, X \leq Y)}{P(X \leq Y)} = 0.117$$

$$(2) F_X(x | X \leq Y) = P(X \leq x, X \leq Y) = \frac{P(X \leq x, X \leq Y)}{P(X \leq Y)} = \frac{\int_0^x \int_u^x f(u, v)dv du + \int_0^x \int_x^1 f(u, v)dv du}{P(X \leq Y)} = \frac{\frac{1}{2}(x^2 + x^3 - x^5)}{0.5} =$$

$$x^2 + x^3 - x^5$$